

Chapter 9: Motion and Types of Motion

Textbook Exercise Solutions & Answers



1. Identify the types of motion.

(a) Movement of the earth around the sun:

periodic, circular

(b) Movement of ceiling fan:

circular

(c) A rocket launched from the ground:

linear

(d) A fish swimming in water:

random

(e) The plucked string of a sitar:..... .

oscillatory motion

2. Fill in the blanks.

(linear, non-linear, uniform linear, non-uniform linear, uniform circular, random, circular, non-uniform circular)

(a) If a ball is released from the terrace of a building, it comes down in motion. On the other hand, it reaches the ground in motion if it is thrown with a force away from the terrace in a direction parallel to the terrace.

uniform linear, non-uniform linear

(b) The motion of an aeroplane on the runway before take off is

linear

(c) The kite looking for its prey flies with motion in the sky.

circular

(d) Children sitting in a rotating giant wheel have motion, while those sitting in a merry-go-round have a motion.

uniform circular, non-uniform circular

3. How are we different?.

(a) Oscillatory motion and Linear motion.

Oscillatory motion	Linear motion
1. In oscillatory motion, body swings back and forth. e.g. Motion of a swing, motion of pendulum of a clock.	1. In linear motion, an object shows displacement in a straight line, e.g. A train in motion, motion of marching soldiers.

(b) Linear motion and Random motion

Linear motion	Random motion
1. Motion in a straight line is called linear motion. e.g. Motion of a train	1. The motion that changes its direction and speed continuously is called random motion, e.g. Motion of a bird.

(c) Random motion and Oscillatory motion

Random motion	Oscillatory motion
1 The motion that changes its direction and speed continuously is called random motion, e.g. Motion of a butterfly	1. The motion of a body that is swinging back and forth is called oscillatory motion, e.g. Pendulum of a clock, the wing of a bird.

4. Explain in your own words, giving one example each.

(a) Linear motion

An object that shows displacement along a straight line is called linear motion, e.g. A vehicle moving on a road.

(b) Oscillatory motion

The motion of a body swinging back and forth is called oscillatory motion, e.g. Motion of a pendulum of a clock.

(c) Circular motion

The motion of an object along a circular path is called circular motion, e.g. Motion of a ceiling fan.

(d) Random motion

The motion that changes its direction and speed continuously is called random motion, e.g. Motion of a butterfly.

(e) Periodic motion

The repetitive motion in which the moving object passes through a certain point again and again after a fixed period is called as periodic motion, e.g. The minute hand of a clock.

5. Answer the following questions in your own words.

(a) Which types of motion are seen in birds flying in the sky?

1. The birds flying in the sky exhibit random motion.
2. The wings of the birds show oscillatory motion.

(b) Write in detail about your experience of various types of motion while riding a bicycle on a road.

1. The cycle itself shows linear motion.
2. The wheels of the cycle show circular motion.
3. The cycle chain shows periodic motion, if the speed is uniform.
4. The handle bar shows random motion.

6. Complete the puzzle using words for types of motion:

Complete the puzzle using the words for types of motion. 1. A spring is stretched and one end is released. 2. A minute hand. 3. A see-saw. 4-5. Children in a march past. 6. A stone rolling down a hillside.

1. Oscillatory
2. Circular
3. Periodic
4. Uniform
5. Linear
6. Random